Address Data Processing and Classification for Validation for the State of Connecticut Master Address Dataset Creation

CT Data Team 2

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1. Project Overview

Objective

The Connecticut GIS office is advancing the development of a comprehensive master address dataset, currently comprising 1.5 to 1.8 million records, sourced from service providers, businesses, and state operations. CT Data Team 2 has completed efforts to streamline the processing, validation, and maintenance of these address-related datasets, ensuring consistency, completeness, and reliability of records. This work now supports and strengthens location identification for a wide range of agencies and sectors. Moreover, the resolution of records with issues related to completeness and accuracy helps minimize inefficiencies in data management moving forward.

Business Problem

Connecticut's address data infrastructure faced persistent maintenance challenges stemming from frequent changes in real estate and infrastructure, as well as inconsistencies caused by integrating multiple administrative sources. Few datasets currently align with recognized standards such as NENA, resulting in structural mismatches, missing fields, and data duplication. These issues complicate critical operations including emergency dispatch, utility management, and inter-agency collaboration. CT Data Team 2 has identified these gaps and underscored the need for a scalable, systematic solution, one that leverages – automation, data science, and clustering methodologies – to strengthen the integrity, usability, and long-term sustainability of the state's address dataset.

Expected Impact

The standardization of address data across Connecticut presents a critical opportunity to improve efficiency, interoperability, and service delivery across multiple sectors. Although this data is currently maintained solely by the Connecticut Department of Emergency Services and Public Protection (DESPP), it is also utilized (often in localized or inconsistent formats) by the Department of Public Health (DPH), the Secretary of the State (SotS), the Department of Transportation (DOT), and numerous regional and local entities. The absence of a consistent statewide standard has resulted in data fragmentation, inefficiencies, and duplicated efforts. CT Data Team 2 has supported efforts to establish a unified and authoritative dataset that strengthens interagency collaboration and data sharing, while also unlocking broader benefits for public safety, economic development, environmental conservation, and public health. This standardized approach further promotes service equity and responsiveness to the public through improved emergency response, streamlined infrastructure planning, and enhanced access to government services.

Key Stakeholders

- State of Connecticut, GIS Office (Office of Policy and Management)
- Department of Emergency Services and Public Protection (DESPP)

Success Criteria and Measurable Outcomes

Success was defined by meeting stakeholder requirements and demonstrating measurable improvements in data quality and consistency, which was achieved through regular follow-up meetings and email communications. One key outcome was ensuring our classification model incorporated human-based evaluations by comparing manual heuristic scoring, based on human-defined weights, against an automated classification model of the same heuristic framework.

To conduct this comparison, our team first used standardization, clustering, and classification methods alongside manual heuristic scoring to define weights for each of the seven engineered features, which can be found in the Feature Engineering section.

Using these features, we applied variable weightings between 10–15% to establish the scoring criteria for a "likely valid", "potentially invalid", and "needs review" address based on human evaluation. After assigning manual classifications, we trained a Random Forest classification model to predict outcomes based on engineered features.

Once both manual and model-based classifications were finalized, we created a confusion matrix to compare predictions and assess model performance. This allowed us to verify that the model was the best fit for the task. In evaluating precision, we aimed for a well-balanced model that was not overfitted and avoided perfect scores (1.0) for F1, precision, and recall. Instead, we targeted for a healthy range between 70–99%.

2. Data Understanding and Preparation

Data Sources and Collection Methods

CT Data Team 2 has utilized 10 datasets provided by the state, along with two publicly available datasets sourced from emergency services (e.g., 911), 2024 parcel address data, utilities, internet service providers, and voter registration. These datasets formed the basis of our data processing and modeling efforts.

Below is more detailed information on the technical aspects of the datasets:

- **Data (Datasets 1-10):**
  - Anonymized utility, broadband internet, and voter registration datasets for residential and commercial addresses in Connecticut. External and private datasets are sourced outside of the GIS office but cannot be accessed without proper permissions.
- **Parcels:**
  - Public dataset provided by GIS for the year 2024. This data set was internally produced by the GIS office and is publicly available.
- **E911:**
  - From CT Department of Emergency Services and Public Protection. This dataset was provided by a source outside of the GIS office and is publicly available.

Data Quality Assessment

Across the datasets used in this project, CT Data Team 2 has identified minor issues affecting data validity, statewide address coverage, record uniqueness, and structural consistency. While only the E911 dataset was expected to conform to the NENA addressing standard, other sources contained address components that deviate from that format. Variations, such as missing leading zeros in ZIP codes, duplicate records, inconsistent column structures, and schema misalignments, were anticipated and did not pose a significant risk to downstream modeling. These findings reinforced the importance of pre-processing and data cleaning to ensure each address is validated as a real-world location and successfully standardized.

Data Preprocessing and Cleaning

To address previously identified quality issues and support alignment with the NENA addressing standard, CT Data Team 2 has applied the following preprocessing and cleaning steps across all datasets as we recognized that full conformance may not apply to every data source. Any datasets added to this analysis in the future should already meet the appropriate formatting requirements or be updated using the same guidelines outlined below. These measures ensure consistency, enable seamless integration, and support accurate identification of real-world locations:

- Column Standardization:
    - Extraneous columns have been removed, and consistent naming conventions applied to streamline schema alignment and simplify downstream processing.
- Null Handling:
    - Null values have been standardized to distinguish between truly missing data and fields that are intentionally blank or non-applicable.
- Fuzzy Matching:
    - Formatting and spelling inconsistencies have been standardized in the Street Pre-Directional, Street Post-Directional, and Street Post-Type columns.

Additionally, the preprocessing framework has been designed to be flexible and standardized. If the existing datasets have changed or a new dataset is incorporated, the preprocessing methods will have to be updated to accommodate this change. The framework serves as a foundation for stakeholders, giving them the ability to make updates as needed.

Feature Engineering

Building on the cleaned and preprocessed data, CT Data Team 2 has introduced a targeted set of engineered features that elevate address-level analysis. These additions consolidate critical signals into a single reasoning layer, enabling more accurate classification, streamlined decision-making, and clearer identification of address strengths and limitations. The following features include:

- Zip_City:
    - Identifies municipalities associated with the provided ZIP Code.
- Zip_Flag:
    - Classifies the validity of a given ZIP Code with three possible categories: Valid CT ZIP Code, Invalid CT ZIP Code, or Not a CT ZIP Code.
- Lat_Long_Present
    - Boolean indicator set to true (1) if latitude and longitude are present in the address record, and false (0) otherwise.
- Muni_in_Zip_City
    - Boolean feature set to true (1) if the municipality in the address is found within the municipalities linked to the corresponding ZIP Code; otherwise, false (0).
- Source_Dataset_Count
    - Returns the number of datasets in which the address appears.
- E911_Presence
    - Boolean feature indicating true (1) if E911 is listed among the source datasets for a given address; otherwise, false (0).

- Validated_Dataset_Count
    - Returns a count (0, 1, or 2) of validated datasets containing the address. For this project, validated datasets include E911 and Dataset 10.
- ResComm_Completeness
    - Boolean flag that checks for the presence of all primary address components: Address Number, Street Name, Street Type, Municipality, and ZIP Code.
- LLAdd_Completeness
    - Boolean flag indicating whether the record likely represents a transformer or generalized location. Set to true (1) if Street Name, Street Type, Municipality, ZIP Code, and latitude/longitude are present, excluding Address Number.

In addition to this, we used binary features to assess:

- Completeness of residential and non-residential addresses
- Presence of geographic coordinates
- Inclusion in the E911 dataset
- Municipality–ZIP Code alignment

It is important to note that records missing critical information from addresses were marked as "needs review." to allow stakeholders to view these addresses more easily. Based on stakeholder feedback the final dataset also includes:

- Source dataset counts
- Confidence scores
- Flagged issues (with classification categories specified as needed)

Lastly, as dataset 10 and the E911 dataset are accurate by assumption, they will be used as our baseline for evaluation. In addition, intersource agreements will be used as a proxy for validity.

Note: *Final CSV outputs do not retain feature columns, but the **Reasoning** column remains to provide interpretability and transparency into the modeling decisions.*

3. Exploratory Data Analysis (EDA)

Descriptive Statistics

Across the 12 datasets used for this project, CT Data Team 2 worked with a total of 6,701,805 address records to support model development. We identified key variables, such as Address Number, Street Name, Municipality, ZIP Code, Latitude and Longitude as essential for defining a unique and complete address across both common and edge-case scenarios.

In datasets 1–10 and E911, the following summary statistics were found:

- Add_Number field was completed in 99.25% of records, demonstrating strong coverage.
- St_Name field showed a completion rate of 99.58%
- Post_Code field was nearly complete with ZIP Code data present in 80.70% of entries.
- Latitude was populated in only 37.82% of records

- Longitude was populated in 37.83%

These statistics indicated a notable gap in geolocation coverage that was factored into our analysis and modeling approach.

Initial Insights

All towns in the state of Connecticut are represented in the final combined dataset, confirming full geographic coverage and statewide inclusivity. When visualized as density maps, address concentrations closely mirror population density, validating the dataset's alignment with demographic distribution.
CT Data Team 2 observed notable variation in how towns, villages, and census-designated places were referenced across the datasets. To address this, we developed a method for identifying synonymous place names. One effective validation approach involved using a trusted ZIP Code database provided by the United States Postal Service to match ZIP Codes to corresponding town and village values.

During preprocessing, we also identified formatting inconsistencies, including ZIP Codes missing leading zeros. The ZIP Code fields were corrected and standardized, and each entry was validated to confirm that it represented a real Connecticut ZIP Code. This was a key indicator of whether an address was legitimate and relevant to the state.

Assumptions & Limitations

CT Data Team 2 has operated under a core assumption: if a records information is present and not flagged for formatting issues, it is considered valid. For instance, if the Street Name is listed as “Dr,” it is accepted as correct, even though it may have been intended as “Terr,” another legitimate value. Given the scale of the dataset, manual verification of each address is not feasible, which presents a known limitation. Additionally, infrastructure changes and shifting political boundaries (e.g. new developments, demolitions, or updates to town or village jurisdictions) introduce further constraints. These dynamic conditions are not captured in static datasets and may affect address accuracy over time.

4. Modeling Approach

Modeling Approaches Explored

1. **Clustering**
    a. Fuzzy Matching
    b. DBSCAN
    c. Greedy Leader Clustering
    d. Graph-Based Clustering
2. **Supervised Classification**
    a. Logistic Regression
    b. Random Forest
    c. XGBoost
    d. Light GBM

Chosen Model Selection & Rationale

1. **Clustering** | Graph-Based Clustering and Blocking
    a. Identify groups of similar addresses, not just pairs. Can help us in cases of conflation and identify cases that may be likely approximate duplicates in need of verification
    b. Useful in handling large datasets by grouping many-to-many relationships and is adaptable to both textual and spatial features we are presented with
2. **Supervised Classification** | Random Forest Classifier Decision Tree Modeling
    a. Enhances clustering by learning feature importance

    b. Provides classification bucketing and subsequent confidence, supporting threshold-based decisions and confidence scoring

Training & Validation Strategy

1. **Data Splitting**
    a. Data was split into training (70%), validation (15%), and test (15%) sets.
2. **Cross Validations**
    a. Compared algorithmically weighted confidence score/classification from dataset to model output to see if similar classification is found.
    b. Agreement between heuristic and supervised model was around 87% (see confusion matrix in final presentation)
3. **Heuristic Scoring (Manual)**

Classification Group	Threshold	Count
Potentially Invalid	0-50	309,360 (5.11%)
Needs Review	51-75	2,938,094 (48.52%)
Likely Valid	76-100	2,808,094 (46.37%)

4. **Performance Metrics**
    a. Summary statistics on cluster size help analyze the likelihood that the clustering model achieved its goal of identifying addresses that were duplicated across multiple source datasets.
    b. Precision, Recall, Accuracy, and F1-Score on the entire dataset after classification modeling.
    c. Confidence Scores are based on manual weightings of created features. This will help us classify addresses with flags of “likely valid”, “needs review”, and “potentially invalid”.

    d. Model confidence that it has successfully classified an address is also output.

Baseline & Alternative Approaches

- **Alternatives Explored:** Fuzzy-Matching and Tree-Based Combination, Clustering and Logistic Regression Combination, Fuzzy-Matching and Logistic Regression Combination

- Selected Baseline: Clustering and Tree-Based Model Combination

5. Evaluation & Preparation

Evaluation Metrics

Models were evaluated using accuracy, precision, recall, and F1 score, with an F1 score baseline target of 80% to balance performance with realistic expectations. Confidence in the supervised classification model was assessed in instances where it aligned with or diverged from manual heuristic labeling. Metric selections were documented to clarify their evaluation functions and the trade-offs they introduced in interpreting results. Ground truth was verified through intersource agreement between manual and supervised classification outputs. Importantly, scores approaching 1.0 were intentionally avoided, as they indicated overfitting, which was an outcome deemed undesirable for this application.

Interpretability

Results are interpreted and communicated through the usage of visualizations, including those for the following metrics:

- Agreement between heuristic and supervised models, in terms of classification distribution
- Modeling confidences by classification bucket and correlated source dataset

Error Reporting

Below are the screenshots of the error report generated by Python code. This report is organized by dataset and documents inaccuracy, accuracy, classification statistics and metrics pertaining to the final dataset.

These reports can be adjusted to be organized by geographic locations (e.g. cities, postal code, etc.) rather than what is currently set as (e.g. source dataset) to gain additional insight.

Source Dataset	Inaccuracies	Accuracies	Classification Stats
E911	• **Normalization Issues**: 0/1168657 (0.0%) • **Invalid Zips**: 0/1168657 (0.0%) • **Conflation**: 1168657/1168657 (100.0%) • **Conflation Average Size**: 7.03 • **Records to Return To Provider**: 19179/1168657 (1.6%)	• **Complete Geospatial Data**: 0/1168657 (0.0%) • **Source of Truth Overlap**: 1168657/1168657 (100.0%) • **Residential/Commercial Address Completeness**: 1149478/1168657 (98.4%) • **Geospatial Address Completeness**: 0/1168657 (0.0%) • **Accurate City Zip Correlation**: 1087788/1168657 (93.1%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 4350 (0.0%) • **Needs Review**: Modeling - 24331, Heuristic - 109559 (22.2%) • **Likely Valid**: Modeling - 1125147, Heuristic - 1054748 (106.7%)
PARCELS	• **Normalization Issues**: 148/1026972 (0.0%) • **Invalid Zips**: 1026972/1026972 (100.0%) • **Conflation**: 998113/1026972 (97.2%) • **Conflation Average Size**: 9.43 • **Records to Return To Provider**: 23382/1026972 (2.3%)	• **Complete Geospatial Data**: 0/1026972 (0.0%) • **Source of Truth Overlap**: 0/1026972 (0.0%) • **Residential/Commercial Address Completeness**: 1003590/1026972 (97.7%) • **Geospatial Address Completeness**: 0/1026972 (0.0%) • **Accurate City Zip Correlation**: 0/1026972 (0.0%)	• **Potentially Invalid**: Modeling - 253974, Heuristic - 274244 (92.6%) • **Needs Review**: Modeling - 749616, Heuristic - 752727 (99.6%) • **Likely Valid**: Modeling - 0, Heuristic - 0 (N/A)
Dataset 1	• **Normalization Issues**: 0/33853 (0.0%) • **Invalid Zips**: 0/33853 (0.0%) • **Conflation**: 33804/33853 (99.9%) • **Conflation Average Size**: 8.06 • **Records to Return To Provider**: 248/33853 (0.7%)	• **Complete Geospatial Data**: 30824/33853 (91.1%) • **Source of Truth Overlap**: 0/33853 (0.0%) • **Residential/Commercial Address Completeness**: 33573/33853 (99.2%) • **Geospatial Address Completeness**: 30609/33853 (90.4%) • **Accurate City Zip Correlation**: 33812/33853 (99.9%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 48 (0.0%) • **Needs Review**: Modeling - 7972, Heuristic - 692 (1152.0%) • **Likely Valid**: Modeling - 25633, Heuristic - 33113 (77.4%)
Dataset 2	• **Normalization Issues**: 1/50171 (0.0%) • **Invalid Zips**: 0/50171 (0.0%) • **Conflation**: 48933/50171 (97.5%) • **Conflation Average Size**: 9.12 • **Records to Return To Provider**: 395/50171 (0.8%)	• **Complete Geospatial Data**: 45537/50171 (90.8%) • **Source of Truth Overlap**: 0/50171 (0.0%) • **Residential/Commercial Address Completeness**: 49718/50171 (99.1%) • **Geospatial Address Completeness**: 45200/50171 (90.1%) • **Accurate City Zip Correlation**: 50107/50171 (99.9%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 85 (0.0%) • **Needs Review**: Modeling - 12987, Heuristic - 1192 (1089.5%) • **Likely Valid**: Modeling - 36789, Heuristic - 48894 (75.2%)
Dataset 3	• **Normalization Issues**: 42/476629 (0.0%) • **Invalid Zips**: 333/476629 (0.1%) • **Conflation**: 460353/476629 (96.6%) • **Conflation Average Size**: 9.37 • **Records to Return To Provider**: 10413/476629 (2.2%)	• **Complete Geospatial Data**: 0/476629 (0.0%) • **Source of Truth Overlap**: 0/476629 (0.0%) • **Residential/Commercial Address Completeness**: 466216/476629 (97.8%) • **Geospatial Address Completeness**: 0/476629 (0.0%) • **Accurate City Zip Correlation**: 474813/476629 (99.6%)	• **Potentially Invalid**: Modeling - 112, Heuristic - 10516 (1.1%) • **Needs Review**: Modeling - 140359, Heuristic - 140262 (100.1%) • **Likely Valid**: Modeling - 325745, Heuristic - 325851 (100.0%)
Dataset 4	• **Normalization Issues**: 0/709569 (0.0%) • **Invalid Zips**: 478/709569 (0.1%) • **Conflation**: 702629/709569 (99.0%) • **Conflation Average Size**: 7.12 • **Records to Return To Provider**: 14891/709569 (2.1%)	• **Complete Geospatial Data**: 0/709569 (0.0%) • **Source of Truth Overlap**: 0/709569 (0.0%) • **Residential/Commercial Address Completeness**: 694678/709569 (97.9%) • **Geospatial Address Completeness**: 0/709569 (0.0%) • **Accurate City Zip Correlation**: 706746/709569 (99.6%)	• **Potentially Invalid**: Modeling - 79, Heuristic - 14902 (0.5%) • **Needs Review**: Modeling - 169580, Heuristic - 146310 (115.9%) • **Likely Valid**: Modeling - 525019, Heuristic - 548357 (95.7%)
Dataset 5	• **Normalization Issues**: 0/163022 (0.0%) • **Invalid Zips**: 18/163022 (0.0%) • **Conflation**: 159224/163022 (97.7%) • **Conflation Average Size**: 7.10 • **Records to Return To Provider**: 1702/163022 (1.0%)	• **Complete Geospatial Data**: 149477/163022 (91.7%) • **Source of Truth Overlap**: 0/163022 (0.0%) • **Residential/Commercial Address Completeness**: 161143/163022 (98.8%) • **Geospatial Address Completeness**: 148138/163022 (90.9%) • **Accurate City Zip Correlation**: 162628/163022 (99.8%)	• **Potentially Invalid**: Modeling - 5, Heuristic - 489 (1.0%) • **Needs Review**: Modeling - 39850, Heuristic - 3754 (1061.5%) • **Likely Valid**: Modeling - 121465, Heuristic - 158779 (76.5%)
Dataset 6	• **Normalization Issues**: 1/208039 (0.0%) • **Invalid Zips**: 134/208039 (0.1%) • **Conflation**: 207877/208039 (99.9%) • **Conflation Average Size**: 7.88 • **Records to Return To Provider**: 8568/208039 (4.1%)	• **Complete Geospatial Data**: 0/208039 (0.0%) • **Source of Truth Overlap**: 0/208039 (0.0%) • **Residential/Commercial Address Completeness**: 199471/208039 (95.9%) • **Geospatial Address Completeness**: 0/208039 (0.0%) • **Accurate City Zip Correlation**: 207855/208039 (99.9%)	• **Potentially Invalid**: Modeling - 12, Heuristic - 8578 (0.1%) • **Needs Review**: Modeling - 48785, Heuristic - 113638 (42.9%) • **Likely Valid**: Modeling - 150674, Heuristic - 85823 (175.6%)
Dataset 7	• **Normalization Issues**: 1/93009 (0.0%) • **Invalid Zips**: 1/93009 (0.0%) • **Conflation**: 91271/93009 (98.1%) • **Conflation Average Size**: 10.49 • **Records to Return To Provider**: 1559/93009 (1.7%)	• **Complete Geospatial Data**: 0/93009 (0.0%) • **Source of Truth Overlap**: 0/93009 (0.0%) • **Residential/Commercial Address Completeness**: 91450/93009 (98.3%) • **Geospatial Address Completeness**: 0/93009 (0.0%) • **Accurate City Zip Correlation**: 92961/93009 (99.9%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 1559 (0.0%) • **Needs Review**: Modeling - 28166, Heuristic - 22529 (125.0%) • **Likely Valid**: Modeling - 63284, Heuristic - 68921 (91.8%)
Dataset 8	• **Normalization Issues**: 16/607618 (0.0%) • **Invalid Zips**: 10/607618 (0.0%) • **Conflation**: 592618/607618 (97.5%) • **Conflation Average Size**: 9.52 • **Records to Return To Provider**: 5992/607618 (1.0%)	• **Complete Geospatial Data**: 0/607618 (0.0%) • **Source of Truth Overlap**: 0/607618 (0.0%) • **Residential/Commercial Address Completeness**: 601616/607618 (99.0%) • **Geospatial Address Completeness**: 0/607618 (0.0%) • **Accurate City Zip Correlation**: 607618/607618 (100.0%)	• **Potentially Invalid**: Modeling - 4, Heuristic - 6002 (0.1%) • **Needs Review**: Modeling - 181903, Heuristic - 587103 (31.0%) • **Likely Valid**: Modeling - 419719, Heuristic - 14513 (2892.0%)
Dataset 9	• **Normalization Issues**: 0/401969 (0.0%) • **Invalid Zips**: 0/401969 (0.0%) • **Conflation**: 390383/401969 (97.1%) • **Conflation Average Size**: 9.34 • **Records to Return To Provider**: 17014/401969 (4.2%)	• **Complete Geospatial Data**: 401969/401969 (100.0%) • **Source of Truth Overlap**: 0/401969 (0.0%) • **Residential/Commercial Address Completeness**: 384674/401969 (95.7%) • **Geospatial Address Completeness**: 384955/401969 (95.8%) • **Accurate City Zip Correlation**: 401969/401969 (100.0%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 2397 (0.0%) • **Needs Review**: Modeling - 116472, Heuristic - 14819 (786.0%) • **Likely Valid**: Modeling - 268483, Heuristic - 384753 (69.8%)
Dataset 10	• **Normalization Issues**: 0/1119945 (0.0%) • **Invalid Zips**: 8027/1119945 (0.7%) • **Conflation**: 1105912/1119945 (98.7%) • **Conflation Average Size**: 7.10 • **Records to Return To Provider**: 11845/1119945 (1.1%)	• **Complete Geospatial Data**: 1119902/1119945 (100.0%) • **Source of Truth Overlap**: 0/1119945 (0.0%) • **Residential/Commercial Address Completeness**: 1096942/1119945 (97.9%) • **Geospatial Address Completeness**: 1107961/1119945 (98.9%) • **Accurate City Zip Correlation**: 1032487/1119945 (92.2%)	• **Potentially Invalid**: Modeling - 940, Heuristic - 2312 (40.7%) • **Needs Review**: Modeling - 25906, Heuristic - 17432 (148.6%) • **Likely Valid**: Modeling - 1081254, Heuristic - 1100201 (98.3%)

Business Impact Assessment

The GIS office was provided with a potential solution for evaluating address validity, one that could be used directly, adapted for specific needs, or referenced as a foundation for further innovation. The development of a systematic approach to cleaning and assessing Connecticut's address-related data resulted in a consolidated, statewide address database. This resource was designed to support state agencies and has the potential to improve operational efficiency and enhance project outcomes.

6. Project Deployment & Timeline

Major Phases, Deliverables & Estimated Completion Dates

Major Phase	Major Phase Deliverables	Completion Date
First Check	- Domain Knowledge - 50% Completion of Exploration Data Analysis	06/06/25
EDA Milestone	- Completion of the Exploratory Data Analysis - Full Understanding of Business Problems and Stakeholder Needs	06/16/25
Second Check	- Fully Cleaned Staging Database - Initial Completion of Model	07/10/25
Modeling Milestone	- Final Completion of Model - Documentation and Code Hand Off	07/24/25

Dependencies

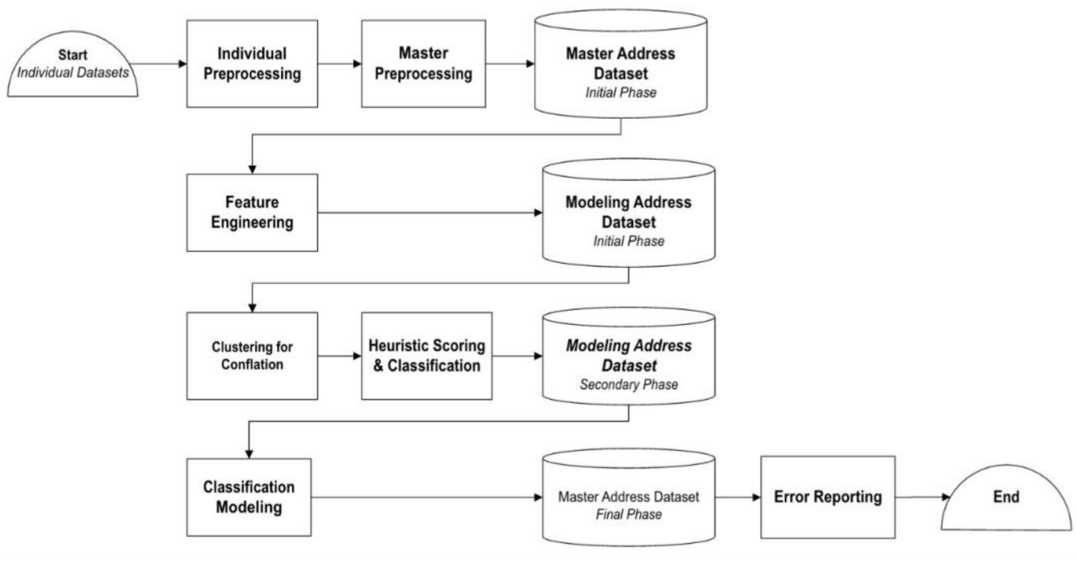
The success of this project relied on a series of structured dependencies to ensure high-quality modeling and evaluation:

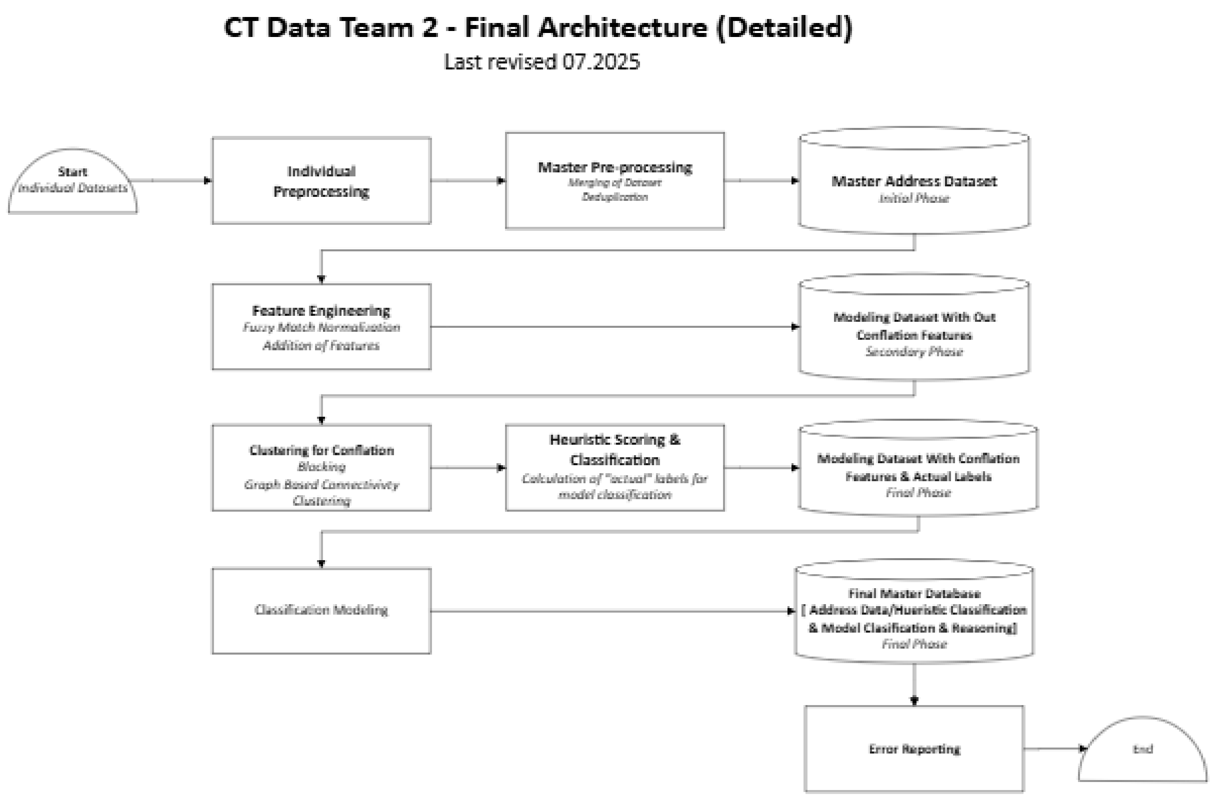
- **Data Standardization**
    - Before any modeling could be initiated, field formatting and data type harmonization were required across all datasets. This process ensured schema alignment and consistency, which served as the foundation for reliable feature engineering.
- **Feature Engineering**
    - Feature construction was contingent on the completion of data cleaning and formatting. These engineered variables provided the signal strength necessary to drive meaningful classification outcomes.
- **Model Development**
    - Sequence Model development was dependent on thorough exploration of statistical confidence approaches. Various model frameworks and combinations were assessed before a final model was selected, ensuring an optimal balance of accuracy, interpretability, and realism.
- **Code Structure & Reusability**
    - Preprocessing code was built to support modularity and scalability. New datasets can be ingested (assuming basic standardization of columns) and processed using existing functions or expanded through custom logic.

These dependencies highlight the steps required to transition from standardized address data to feature and modeling enriched Master Address dataset. Each component built upon its predecessor, ensuring that downstream processes operate on a foundation of technical integrity and contextual relevance.

Architecture

Condensed and Detailed Versions





7. Risk Assessment

Potential Technical Challenges & Concerns

- **Missing Data**
  - Caused difficulties when conflating data and impacted the validity of our model results.
- **Assumption of Data**
  - We may have made incorrect assumptions of the data which can lead to modeling problems.
- **Future Data**
  - Current preprocessing is designed based on data at hand. We cannot know how future data will be formatted.

Resource Constraints

Limited time to develop comprehensive solutions that can accommodate new data and multiple solutions
- Limited use of applications (Geocoder, Google Collaborate). Certain datasets are private and cannot be uploaded to incorporate said applications. Anonymization and
- Budget constraints limit potential use of paid proprietary methods

Mitigation Strategies

CT Data Team 2 employed a range of mitigation strategies to proactively manage risks and ensure successful project delivery:

- **Maintained Open Communication**
  - We established an ongoing line of communication with stakeholders to stay aligned with their priorities and adapt to evolving goals throughout the project.
- **Applied Iterative Flexibility**
  - An agile and iterative development approach allowed us to accommodate changes and integrate stakeholder feedback efficiently at each phase.
- **Upheld Data Security and Process Integrity**
  - All activities adhered to established security protocols to protect sensitive information. We also ensured that all processes were safely repeatable, promoting reliability and compliance.
- **Leveraged Free and Open Source Tools**
  - To address budget limitations, we utilized open-source technologies that allowed for robust functionality without compromising project quality or scalability.
- **Prioritized Procedurally Under Time Constraints**
  - With limited time, we followed a structured prioritization strategy, focusing on the most impactful tasks first to maximize progress and meet key deadlines.

CT Data Team 2

End to End Summary

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Contents

Key Findings

	Preprocessing	Feature Engineering	Confidence Scoring	Clustering	Classification
Purpose	Initial cleaning of datasets and reconciliation to desired schema.	Creates additional features to act as a guideline for future confidence scoring.	Determines the bucket categories for address dataset based on manual scoring.	Clustering techniques to conflate address datasets.	Determines the bucket categories for address dataset based on classification model.
Reasoning	Addresses standardization of the dataset.	Provides a human baseline of features to consider when addressing validity.	Provides a baseline of address validity to compare classification model.	Fills in missing values of addresses to ensure accuracy and completion.	Flags addresses into "Needs Review", "Potentially Invalid" and "Likely Valid"
Stats	***Completeness*** Add Number : 99.1% Post Code: 80.7% Municipality: 84.7% Street Name: 78.8% Unit: 54.5% Everything else: <50%	8 Additional Features (e.g Presence of Lat/Long, Dataset Presence, etc)	**Potentially Invalid** (5.11%) **Needs Review** (48.52%) **Likely Valid** (46.37%)	**Cluster** **Mean size:** 3.11 **Median size:** 3	**Precision:** 90.22%, **Accuracy:** 86.97%, **Recall:** 88.23% **F1-Score:** 88.94%

Business Objectives

Standardized Methods

- Assess address accuracy enhances the reliability and usability of this dataset, enabling government and private entities to deliver critical, location-based services and information more effectively to the community.

Classification of Addresses

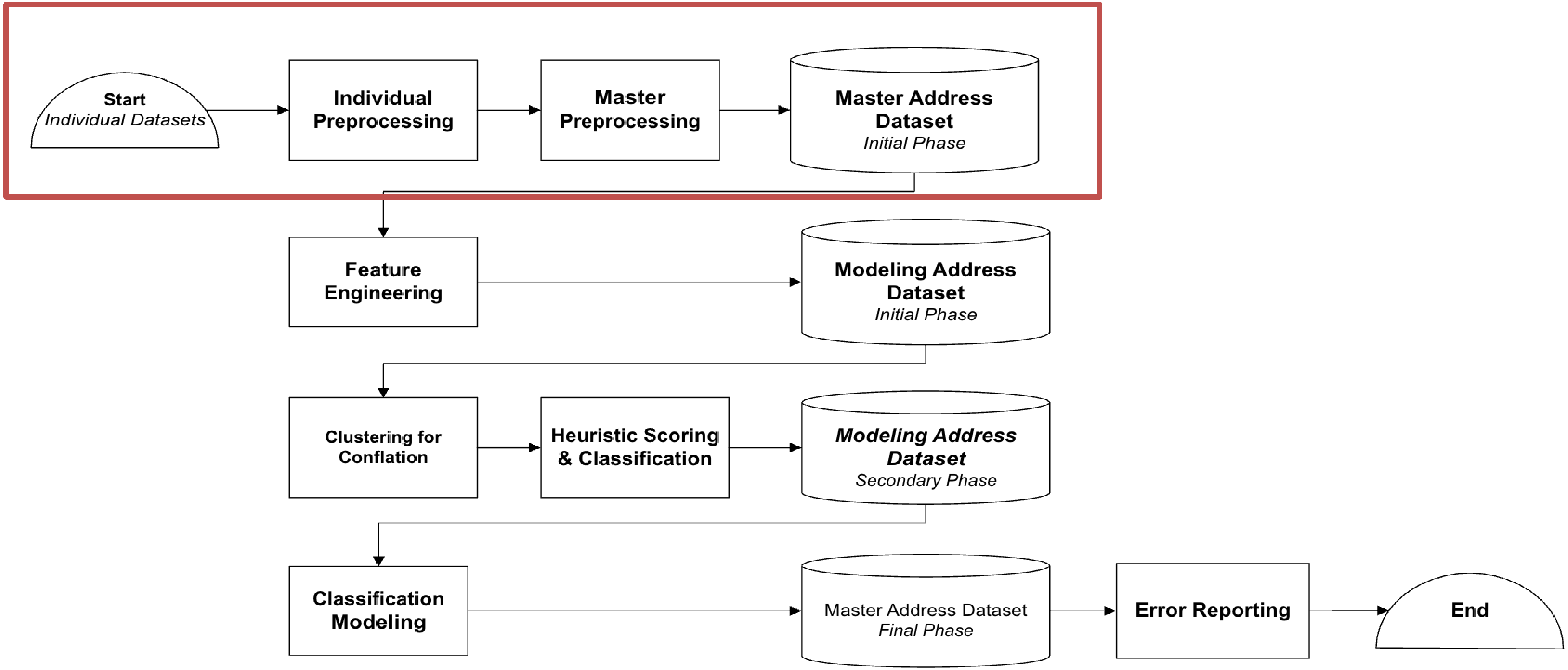
- Classifying addresses into categories like ‘Needs Review,’ ‘Likely Valid,’ and ‘Potentially Invalid’ helps stakeholders determine which addresses are reliable and which require further verification

Error Reporting

- Flagged 'Needs Review' addresses will include detailed reasoning and completion requirements, equipping stakeholders with insights into address quality issues and what’s needed to elevate trustworthiness.

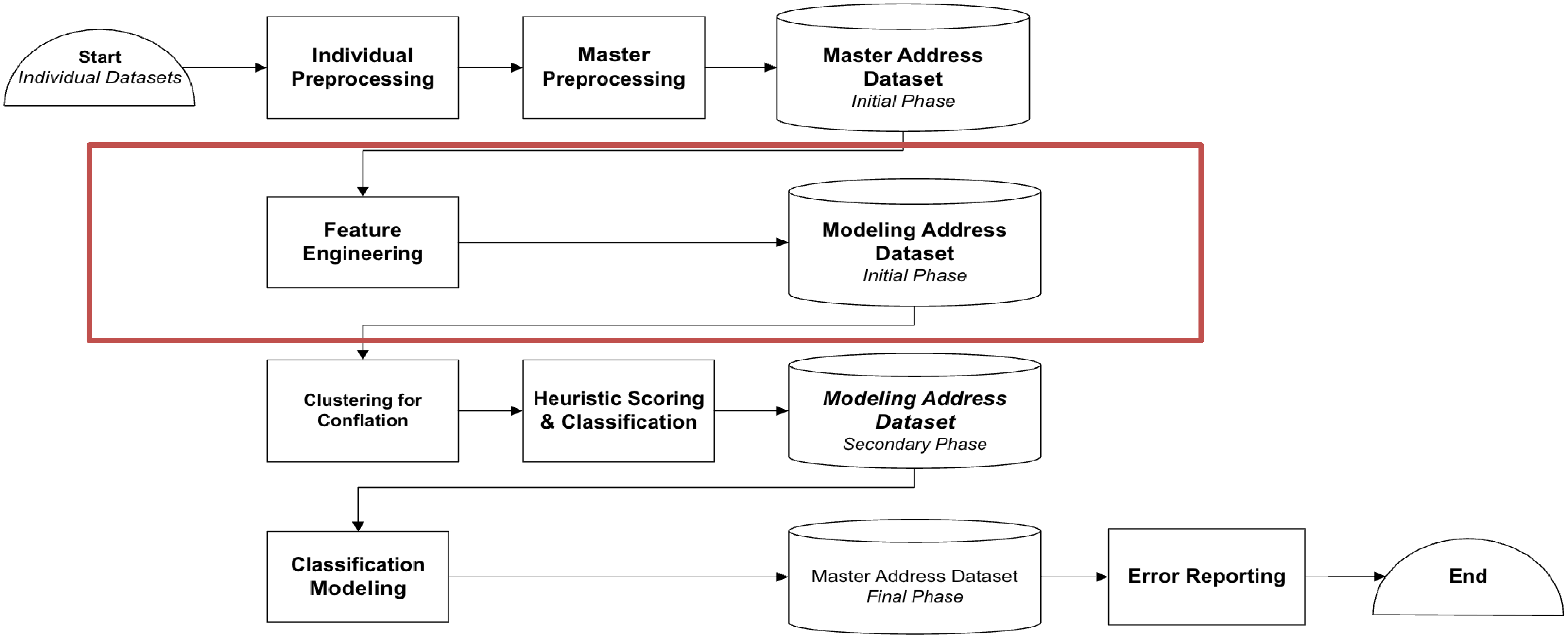
CT Data Team 2 - Final Architecture

Last revised 07.2025



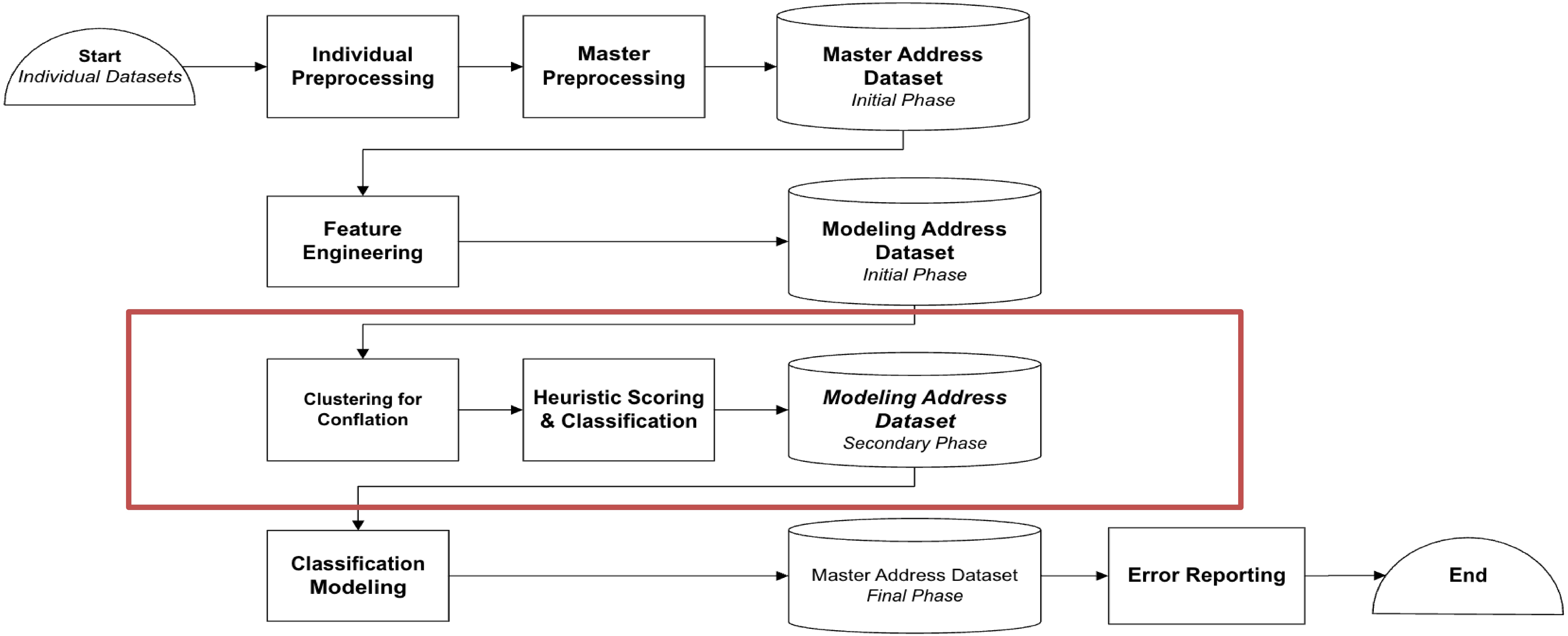
CT Data Team 2 - Final Architecture

Last revised 07.2025



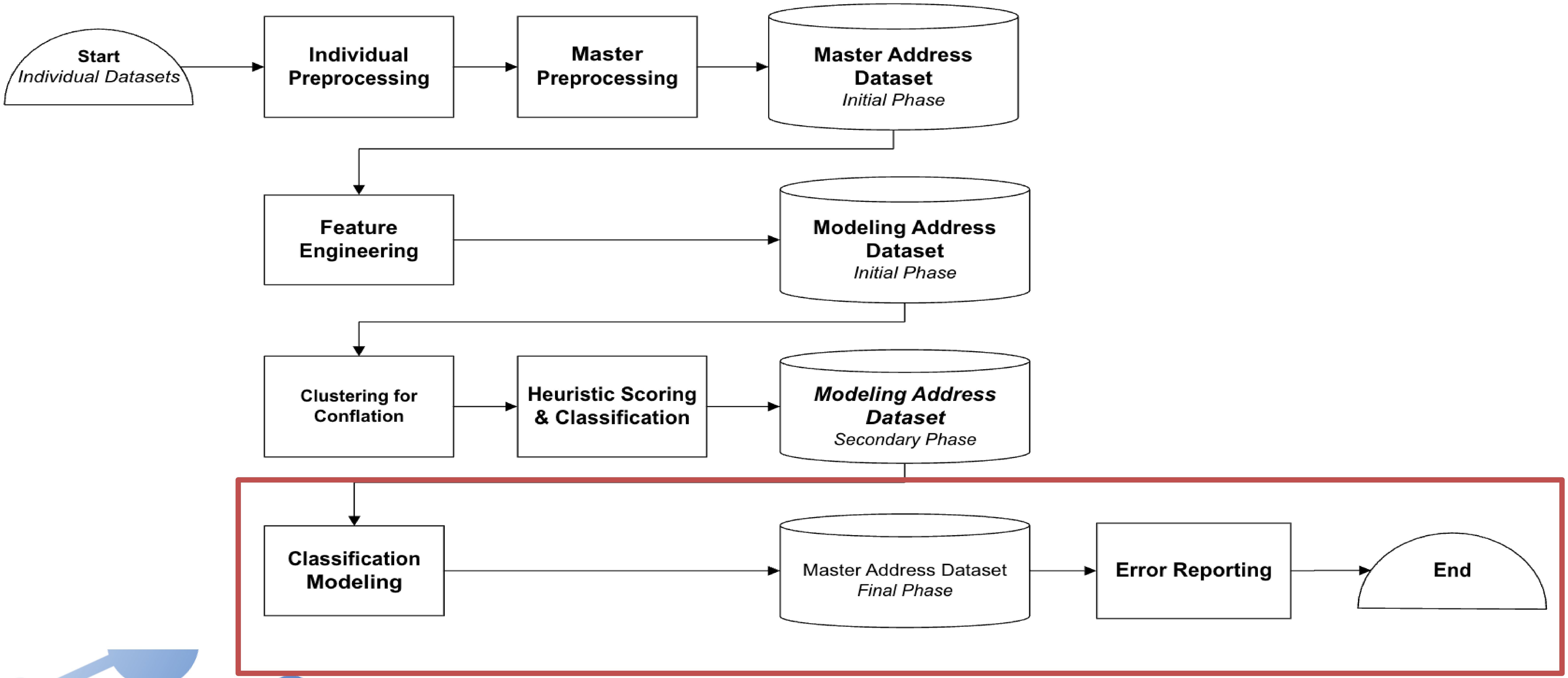
CT Data Team 2 - Final Architecture

Last revised 07.2025



CT Data Team 2 - Final Architecture

Last revised 07.2025



Preprocessing Methods

Standardization

- All provided datasets (**Datasets 1-10, E911 and Parcels**) were modified to conform to the desired schema.
- Fuzzy matching was used to standardize inconsistencies in data formatting (**ex: North vs. N, Drive vs Dr.**)
- Pipeline is designed to take in addresses in **NENA Standard**, so this should be kept in mind if datasets are added in the future

Feature Engineering

- Features were created to allow for more robust understanding of the data, such as the validity of a ZIP code or the completeness of an address
- Used in the manual confidence scoring process to assess address validity

Confidence Scoring Process

Manual Confidence Scoring

- Heuristic, weight-based scoring
- 3 classifications: Potentially Invalid, Needs Review, Likely Valid

123 Main Street, Cromwell, CT 06416, 41.6150, 72.6600

Zip_Flag	Lat_Long_ Present	Muni_in_ ZipCity	E911_ Presence	Dataset_ Presence_Count	Validated_ Dataset_Count	ResComm_ Completeness	LLAdd_ Completeness	Cluster present	Cluster Size	Return to Provider?
✓	✓	✓	✓	2	1	✓	✓	✗	0	✗

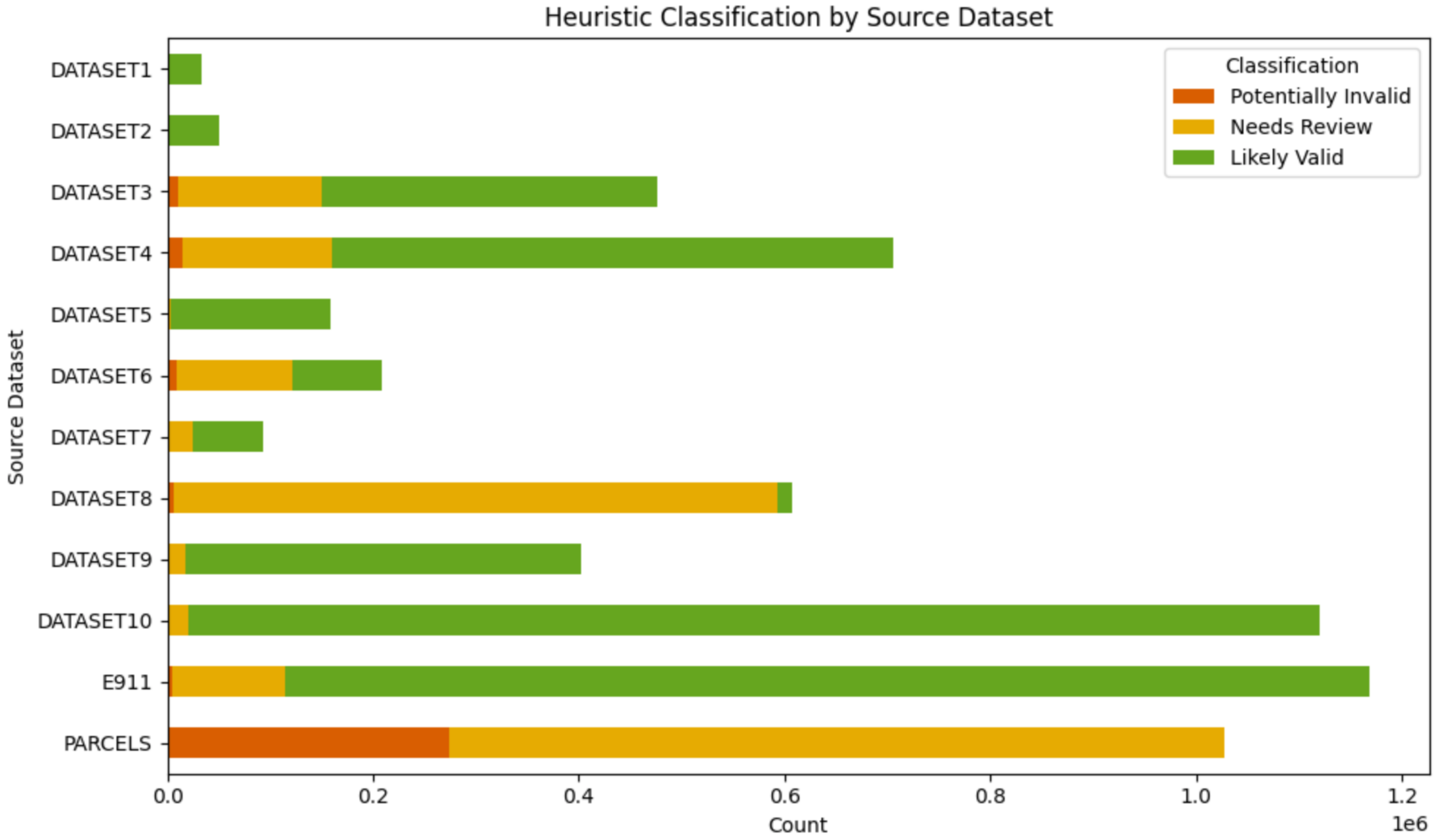
Confidence Score	Heuristic Classification
89.54%	Likely Valid

Confidence Scoring Results

Classification Group	Threshold	Count
Potentially Invalid	0-50	309,360 (5.11%)
Needs Review	51-75	2,938,094 (48.52%)
Likely Valid	76-100	2,808,094 (46.37%)

Confidence Scoring by Dataset

- Key Points:
  - Potentially Invalid makes up a very small portion of overall addresses
  - Lack of "Likely Valid" for Parcels is likely due to missing ZIP Code
  - Validated Datasets, Dataset 10 and E911, make up a large proportion of the likely valid addresses



Clustering for Conflation

Graph-based clustering allows eventual conflation from 6M to 1.6M addresses

Explanation of Final Model

- Graph-based clustering analyzes all address fields and features
- Uses string similarity metrics to "connect" similar addresses
- Outputs a cluster ID related cluster size for EACH of the 6M rows, for eventual conflation

Output and Analysis of Final Model

- **6M** rows assigned to one of **1.6M** clusters
  - 1.5M-1.8M was the **expected** range of final addresses
- **Mean cluster size** = 3.11, **median cluster size** = 3
- Coincides with logic surrounding "Dataset Presence Count"
  - An address appears on average in 3-4 different source datasets

Clustering for Conflation

- Majority of cluster sizes are between 1 and 3
- Cluster size only considered an error if > 12 (Max # of Datasets)
  - Very few cases where this occurs, useful for error analysis

Classification Models Considered

Model	Findings
Logistic Regression	Overly simplistic, too confident in classifications
Random Forest	Overfit, hyperparameter tuning insufficient for fixing this issue
XGBoost	Performed well, hyperparameter tuning improved overfitting
★ LightGBM	Best performing model on our data compared to other models tested

LightGBM Classification Report

--- Predicted Class Distribution ---

Likely Valid: 4,141,157 (69.71%)

Needs Review: 1,544,284 (26.00%)

Potentially Invalid: 255,126 (4.29%)

--- Classification Report ---

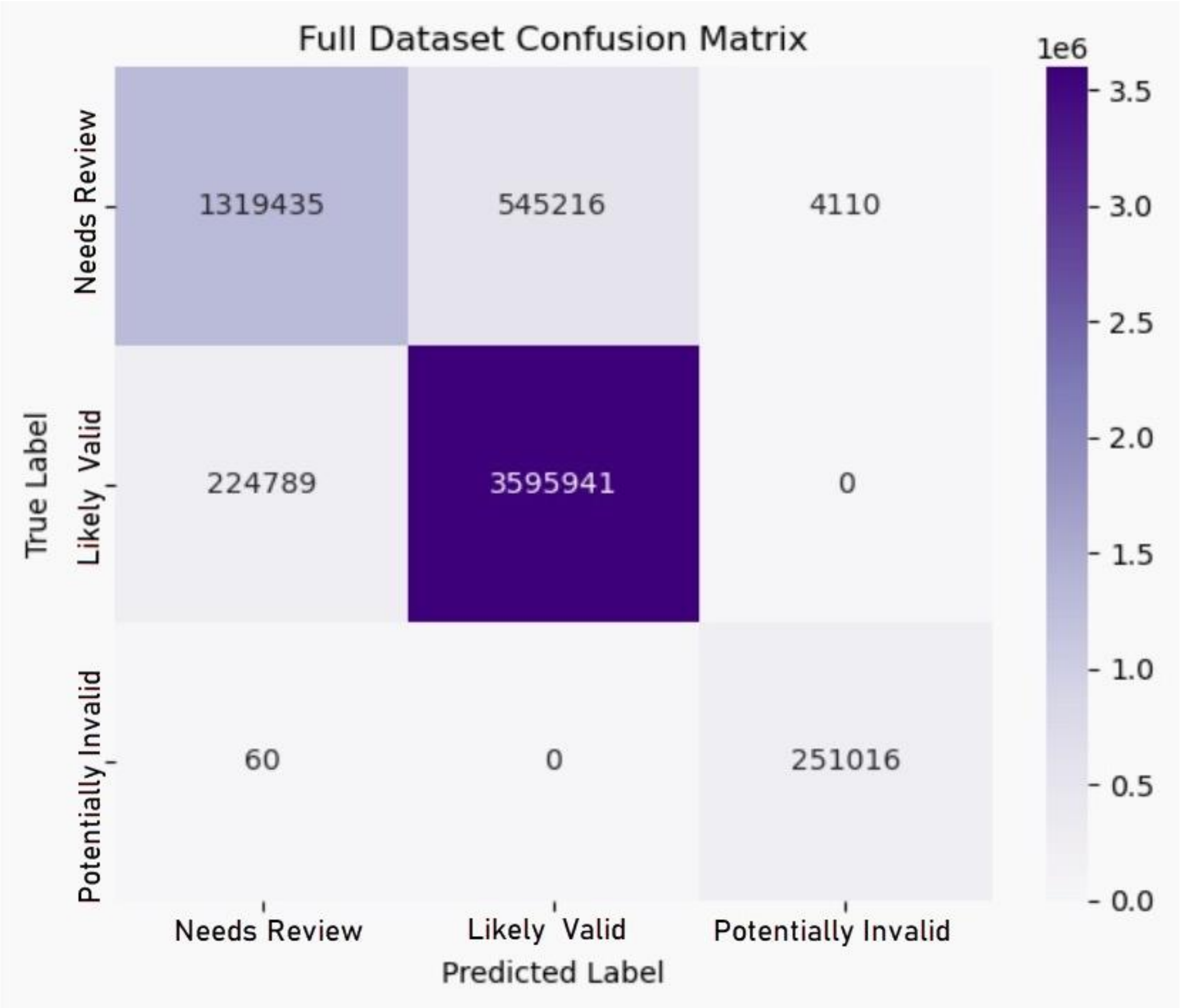
	precision	recall	f1-score	support
Needs Review	0.8544	0.7060	0.7732	1868761
Likely Valid	0.8683	0.9412	0.9033	3820730
Potentially Invalid	0.9839	0.9998	0.9918	251076
accuracy			0.8697	5940567
macro avg	0.9022	0.8823	0.8894	5940567
weighted avg	0.8688	0.8697	0.8661	5940567

Classification Confusion Matrix

- Strong agreement between model and heuristic classifiers

- No egregious disagreements (Likely Valid -> Potentially Invalid)

- Overall agreement: 87%



Results Summary

Standardization

- Consistency: Standardized columns
- Normalization: Fuzzy Matching

Conflation

- Conflation Applicable or Not: Cluster ID marks if Clustering applies
- Level of Conflation: Cluster_Size marks how many records apply to Cluster

Classification

- Two methods of classification for Intersource agreement
  - Heuristic Classification => Human Reasoning Based Formula for Classifying => Serves as Comparison
  - Modeling Classification => LightGBM Classifier

Error Analysis Report

Source Dataset	Inaccuracies	Accuracies	Classification Stats
E911	• **Normalization Issues**: 0/1168657 (0.0%) • **Invalid Zips**: 0/1168657 (0.0%) • **Conflation**: 1168657/1168657 (100.0%) • **Conflation Average Size**: 7.03 • **Records to Return To Provider**: 19179/1168657 (1.6%)	• **Complete Geospatial Data**: 0/1168657 (0.0%) • **Source of Truth Overlap**: 1168657/1168657 (100.0%) • **Residential/Commercial Address Completeness**: 1149478/1168657 (98.4%) • **Geospatial Address Completeness**: 0/1168657 (0.0%) • **Accurate City Zip Correlation**: 1087788/1168657 (93.1%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 4350 (0.0%) • **Needs Review**: Modeling - 24331, Heuristic - 109559 (22.2%) • **Likely Valid**: Modeling - 1125147, Heuristic - 1054748 (106.7%)
PARCELS	• **Normalization Issues**: 148/1026972 (0.0%) • **Invalid Zips**: 1026972/1026972 (100.0%) • **Conflation**: 998113/1026972 (97.2%) • **Conflation Average Size**: 9.43 • **Records to Return To Provider**: 23382/1026972 (2.3%)	• **Complete Geospatial Data**: 0/1026972 (0.0%) • **Source of Truth Overlap**: 0/1026972 (0.0%) • **Residential/Commercial Address Completeness**: 1003590/1026972 (97.7%) • **Geospatial Address Completeness**: 0/1026972 (0.0%) • **Accurate City Zip Correlation**: 0/1026972 (0.0%)	• **Potentially Invalid**: Modeling - 253974, Heuristic - 274244 (92.6%) • **Needs Review**: Modeling - 749616, Heuristic - 752727 (99.6%) • **Likely Valid**: Modeling - 0, Heuristic - 0 (N/A)
Dataset 1	• **Normalization Issues**: 0/33853 (0.0%) • **Invalid Zips**: 0/33853 (0.0%) • **Conflation**: 33804/33853 (99.9%) • **Conflation Average Size**: 8.06 • **Records to Return To Provider**: 248/33853 (0.7%)	• **Complete Geospatial Data**: 30824/33853 (91.1%) • **Source of Truth Overlap**: 0/33853 (0.0%) • **Residential/Commercial Address Completeness**: 33573/33853 (99.2%) • **Geospatial Address Completeness**: 30609/33853 (90.4%) • **Accurate City Zip Correlation**: 33812/33853 (99.9%)	• **Potentially Invalid**: Modeling - 0, Heuristic - 48 (0.0%) • **Needs Review**: Modeling - 7972, Heuristic - 692 (1152.0%) • **Likely Valid**: Modeling - 25633, Heuristic - 33113 (77.4%)

Hand Off Resources

01 High-Level Documentation Final
- Outlines all steps of process
- Includes tried-&-tested methods and reasoning

02 Final Documented Jupyter Notebook
- Full code with comments to indicate functionality
- Commented-out sections for optional csv prints/additional future suggestions

03 Data Dictionary
- All columns including dropped are outlined

04 Error Reporting
- Inaccuracies, accuracies, and classifications statistics per Dataset

05 Final CSV
- Master CSV with all address data with classification columns